EntitleLeave Tracker System

CPIT250 – System Analysis and Design

Prepared by Group 4 (BAR):

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Abstract

The “Entitle Leave Tracker System” is a web-based system designed to help the HR department manage employee leave, contracts, and attendance in an easier and more accurate way instead of doing it manually. The system includes smart features like automatic notifications, leave balance calculation, and attendance tracking. The goal of this project is to reduce errors and improve HR work efficiency.

Introduction

The company has had several issues because of Manual work in the HR department such as delays in leave approvals, forgetting contract renewals, and difficulty tracking leave and attendance. HR staff often rely on Excel sheets and paper-based systems, which are time-consuming and lead to human errors. And when they are managing many employees, this becomes even more difficult and increases the chance of missing important dates or losing track of vital employee data.

Instead of managing employee leave, contracts, and attendance by hand, the HR department could do it more easily and accurately with the help of the web-based “EntitleLeave Tracker System” it is a system designed to provide automatic notifications, leave balance calculation, and attendance tracking. The goal of our project is to reduce human-made errors and help improve the HR staff's work efficiency.

By using this system ,HR staff can save time ,reduce paperwork ,and make faster decisions . Employees also benefit by receiving automatic updates about their leave requests and contract status .Our goal is to make the HR process easier ,faster ,and more accurate through digital transformation .

Problem Description

Managing employees' leaves, contracts, and attendance is a major difficulty for the HR department. The existing system's heavy reliance on manual procedures results in challenges:

- Due to manual processes, leave entitlement tracking can frequently cause errors and delays in approval process.
- Expirations of contracts that go unrecognized could lead to operational issues and legal complications.
- Attendance errors, which may affect employee's performance and reporting.
- Payroll data are liable to inconsistencies and human mistakes.
- No automated alerts for contract expirations/renewals, attendance issues, or leave approvals.
- Data retrieval is inefficient because records are kept in Excel without a centralized structure.

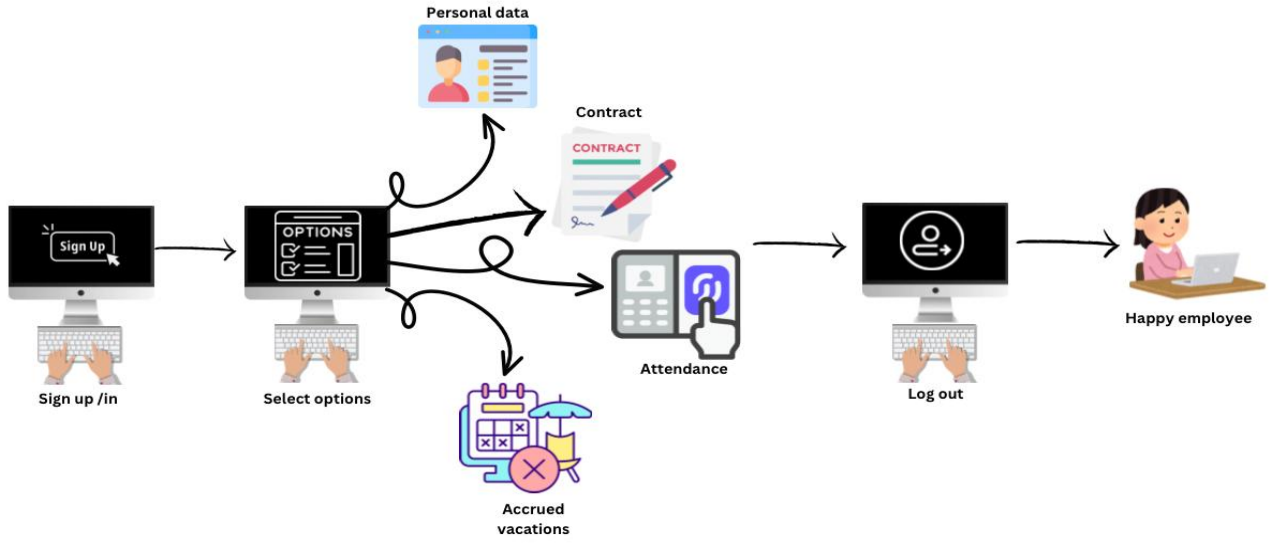
This problem aims to create a system that reduces manual caused errors and increases efficiency by automating contract administration, leave tracking, and attendance monitoring.

Proposed Solution

The Entitled Leave Tracker System is a web platform that meets the needs of the system's main users who are HR employees. It should have the following features:

- **Calculation of Automatic Leave Entitlement:** The system calculates leave balances according to previous leaves and suggests for approval or denial.
- **Automatic Contract Updates:** The system makes sure that employment contracts are updated instantly.
- **Attendance Reports:** Detailed attendance reports are generated which supports performance of employees based on accurate data.
- **Payroll Calculations:** The system automatically saves and calculates payroll data based on contract and up-to-date attendance.
- **Automated Notifications:** Employees receive alerts about contract expirations/renewals, attendance updates, and leave permissions.
- **Centralized Employee database:** An integrated system that contains all of employee's records, including work history, personal information, and performance indicators.

Architecture diagram :



Interview Outline

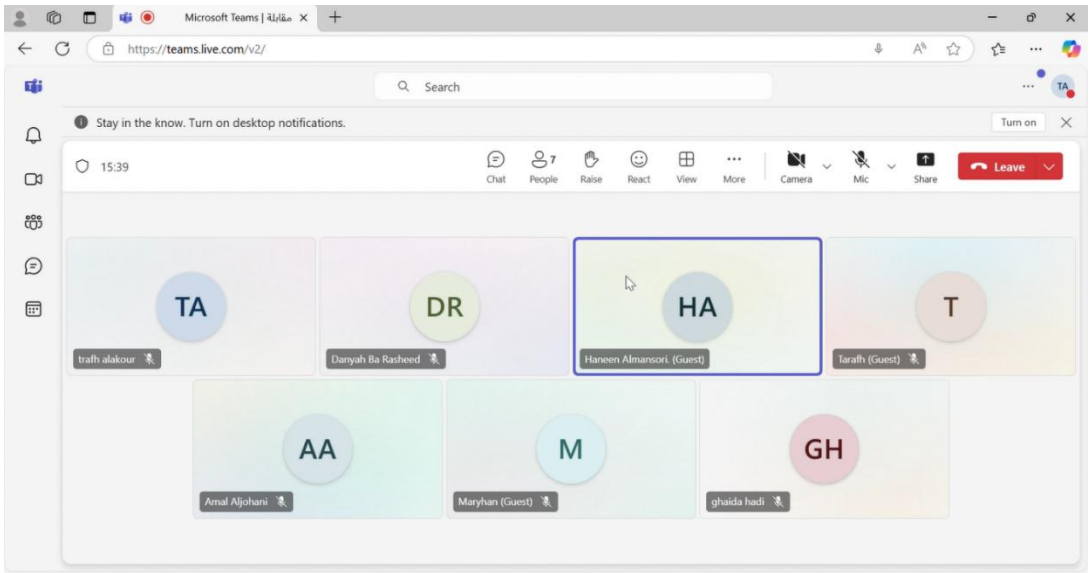
Field	Details
Interviewee	HR \ Haneen Almansouri
Company	Al-Qafila & Al-Yanabee Industrial Advertising Boards
Location	Jeddah, Second Industrial City
Industry	Manufacturing advertising boards for major companies such as Al-Tazaj, Al-Ahli Bank, KFC, Hardee's, Noon, Panda, and more, including street signage.
Interviewer	Tarfaa Alakour, Danyah Ba Rasheed, Ghaida Hadi, Amal Al Juhani, Mirihan Abu Alala
Medium	Online on teams
Start Time	5:00 PM
End Time	5:28 PM

Objectives:

Objective Number	Description
1	Understand HR challenges in managing employee leave, contracts, and attendance.
2	Identify pain points in employee record tracking and update mechanisms.
3	Gather insights on desired system features for automation.

Agenda:

Agenda Item	Duration	Description
Introduction	2 min	Explanation of the project and purpose of the interview.
Background on Project	3 min	Overview of the Leave Tracker System and its benefits for HR operations.
Overview of Interview	2 min	Discussion on topics to be covered and expected outcomes.
Interview Topics & Questions	15 min	HR daily tasks, challenges in employee data management, attendance and leave tracking difficulties, contract management issues, desired automation features.
Summary of Major Points	3 min	Recap of key insights and challenges discussed.
Questions from Interviewee	2 min	Address any queries or additional inputs from the interviewee.
Closing	1 min	Thanking the interviewee and explaining the next steps.



Interview Questions & Responses

Question	Response
What are the main daily tasks handled by the HR department?	The HR department takes care of employees both inside and outside the organization, including recruitment, medical insurance, interviews, attendance tracking, and government affairs management.
How are employee data managed? Is there a clear system in place?	Employee data is stored in an Excel database. Previously, there was no unified system, and efforts were made to manually consolidate the data.
Do you rely on electronic systems, or is most of the work manual?	75% of the work is electronic, with limited manual processes for tasks such as leave and permission requests.
What information is required when hiring a Saudi vs. a foreign employee?	**Saudi:** CV, national ID, certificates, previous experience. **Foreign:** Residency details, sponsorship, visas.
Do you face difficulties retrieving employee data?	Retrieving data is not a problem, but keeping it updated and tracking contract expiration dates is challenging.
Are there any tasks that take too much time?	Calculating leave entitlements is time-consuming as the current system is paper-based and inefficient.
Do errors frequently occur in your work?	Yes, errors occur due to reliance on memory and paper-based systems, especially in leave calculations.
Is it difficult to track attendance, leave, or salaries?	Yes, tracking depends on a single employee. If they are absent, confusion arises. There was an attempt to integrate attendance tracking into a system, but it was not completed.
Are there reports or processes you wish were automated?	Yes, leave, attendance tracking, and employee data updates should be automated.

If a leave management system is built, what features should it have?	It should include notifications for contract expirations and leave, along with automatic employee data updates.
If a contract management system is built, what features should it have?	A comprehensive database with employee details, automatic contract updates, and advance notifications for expiring contracts and residency permits.
Do you face difficulties in tracking attendance?	Yes, the system depends on a single employee. If they leave, confusion arises. There was an attempt to integrate attendance tracking with biometric devices, but it was unsuccessful.
Is Excel your primary tool for managing employee data?	Yes, Excel is widely used for employee and contract data, but there is a need for more advanced solutions.
What tasks take too long and need improvement?	Calculating leave entitlements and tracking contract renewals need improvement as they are manually handled and time-consuming.
If a new electronic system is introduced, what features should it include?	Automatic data updates, contract and leave notifications, a unified employee database, and quick data retrieval.
What challenges do you face in updating employee records?	The absence of an automatic update system makes it difficult to track expired contracts and pending entitlements.
Is there a need for an internal notification system to alert employees about entitlements and leave?	Yes, an internal notification system is essential for alerting employees and managers about leave, contracts, and entitlements.
Is there a difference in handling data for Saudi and foreign employees?	Yes, foreign employees require additional procedures, such as sponsorship transfers and residency renewals.
Is there an intention to adopt a more advanced system instead of Excel?	Yes, there is a desire to transition to a more advanced system, especially for contract and leave management.

Do you face difficulties retrieving employee information when needed?	Yes, as searching for information in Excel is manual, and updates are not always consistent.
Do you need automated reports to track employee performance?	Yes, performance reports are very useful, particularly for attendance, leave, and entitlements.
What challenges do you face in contract updates?	The lack of a contract expiration notification system makes it difficult to track expiring contracts in time.

Requirements:

Functional Requirements:

1. Employee Management:
    - A database system to store, update, and simply retrieve employee information.
2. Leave Management:
    - A tool to process employees' leave requests, and determine which employee is entitled to leave, also the system notifies both employees and managers about leave approvals and the remaining leave balances.
3. Attendance Tracking:
    - Automatic system that's connected to biometric devices such as fingerprint scanners, or face recognition software. The device helps in tracking when the employee arrives, leaves, or if they are late, and if they are absent. Finally, the system generates daily reports for HR and management departments.
4. Contract Management:
    - A specific part of the database holds employees' contracts, sends reminders to the HR department when a contract is about to expire, and oversees the contract renewal process.
5. Reporting & Analysis:
    - Generates automated reports on attendance, leave entitlements, and expiring contracts as well as employee performance reports.

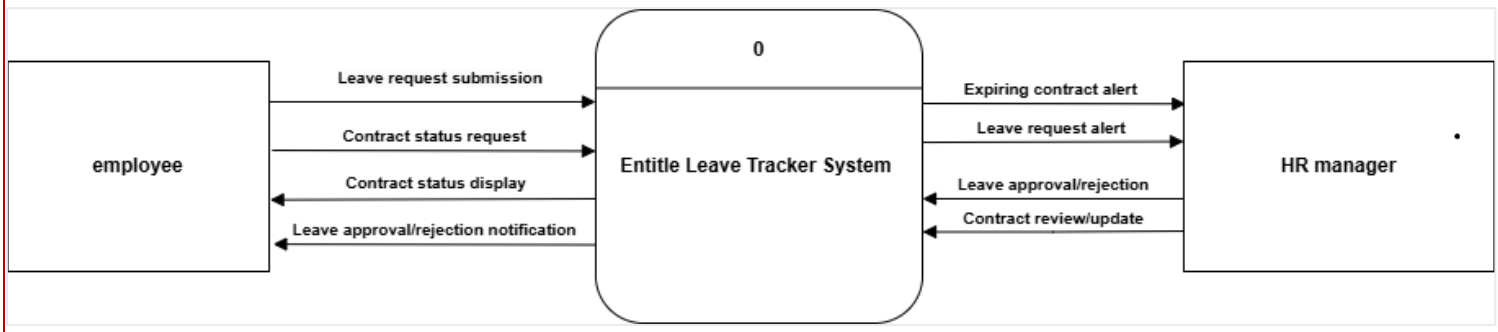
Non-Functional Requirements:

1. System Usability:
    - The system must be simple to use, with a user-friendly interface that doesn't need much training. It should work well on various devices.
2. Performance:
    - The system must be fast, for efficient resource usage and overall productivity. Finally, the system needs to be designed to grow as the company grows.
3. Security:
    - It's of the utmost importance that the company secures employee data with encryption, protects employee login methods, and performs regular system backups to prevent data loss.
4. Reliability & Maintenance:
    - the system should keep a low downtime, it should handle errors efficiently and allow frequent system updates without disturbing daily operations.

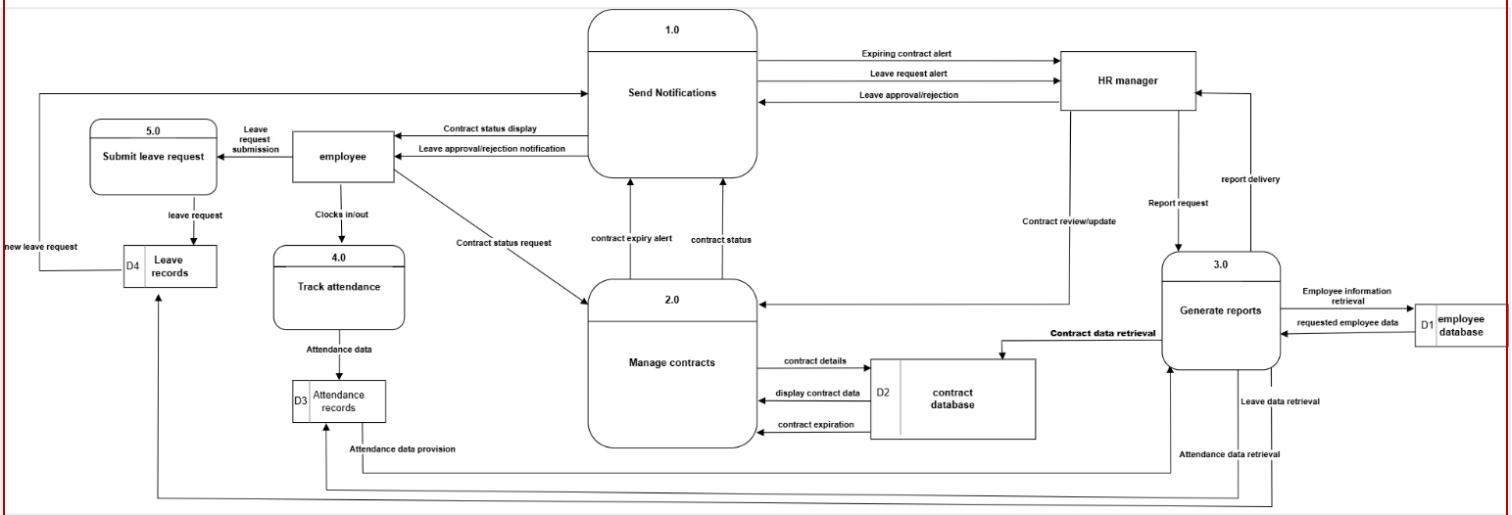
Structuring System Requirements

Data Flow Diagram (DFD):

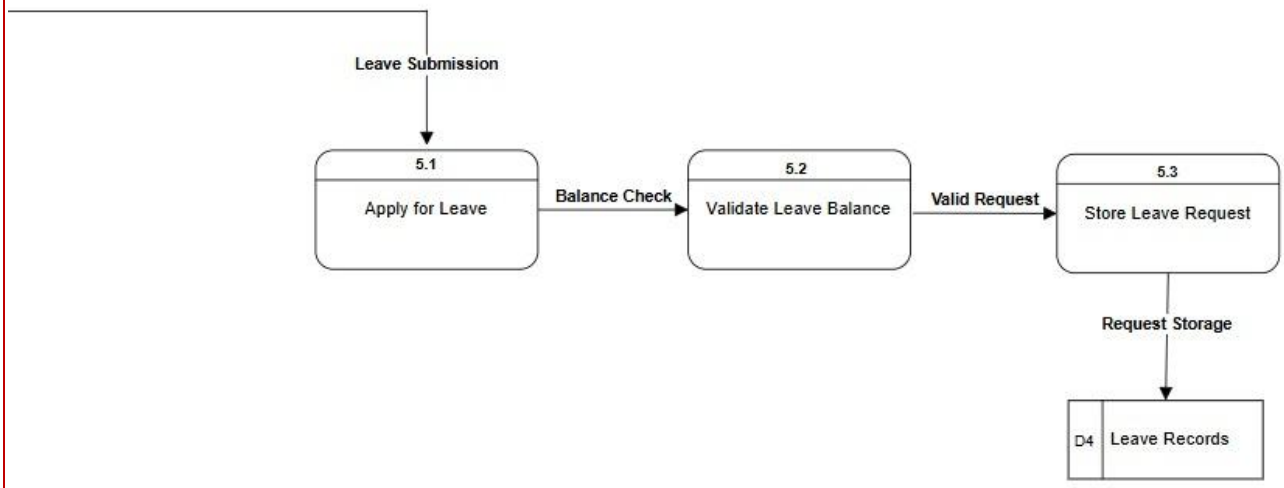
1. Context Diagram



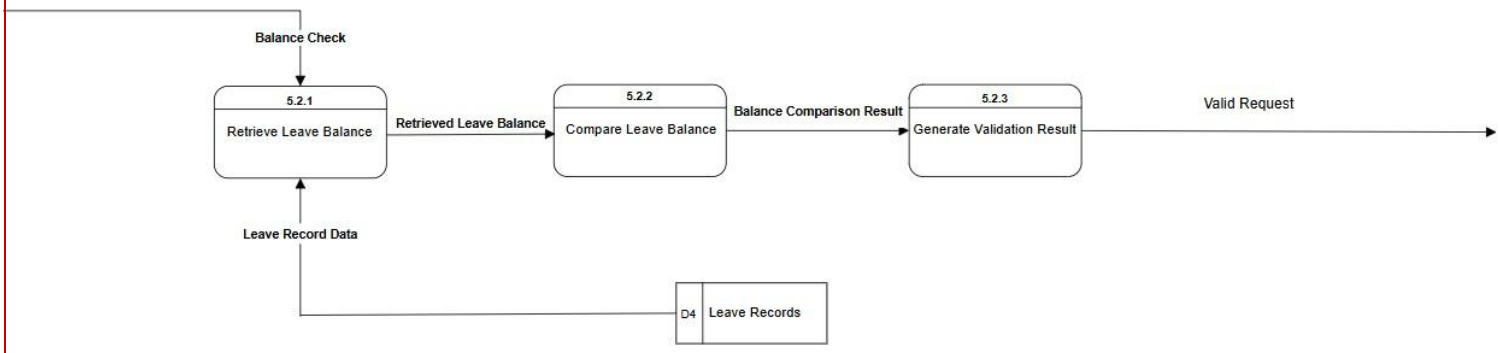
2. Level -0- Diagram



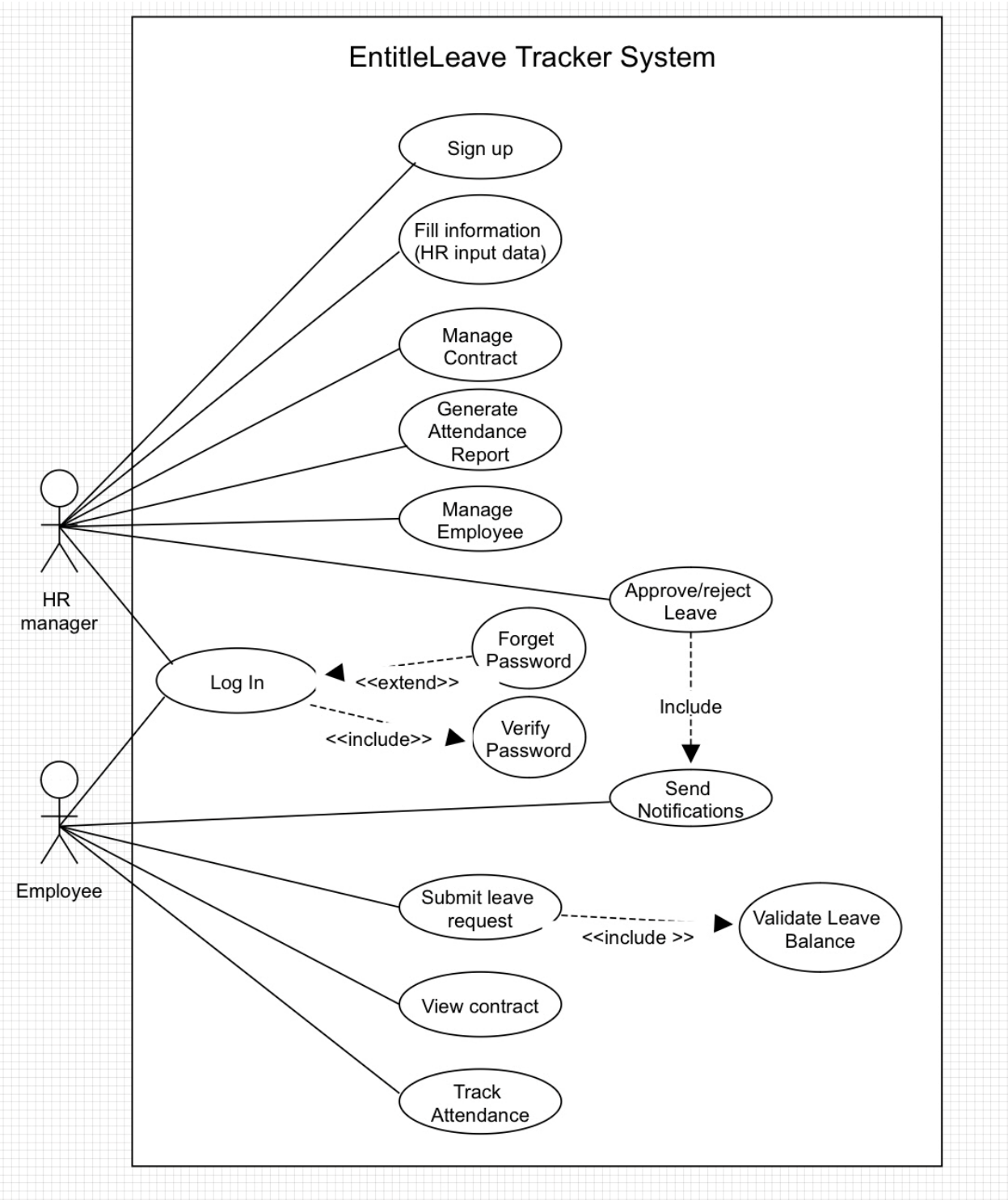
Level -1- of Process 5.0 (Submit Leave Request)



Level 2 of process 5.2 (Validate Leave Balance)



Use Case:


EntitleLeave Tracker System
Sign up
Fill information (HR input data)
Manage Contract
Generate Attendance Report
Manage Employee
Approve/reject Leave
HR manager
Forget Password
Log In
<<extend>>
Include
<<include>>
Verify Password
Send Notifications
Employee
Submit leave request
<<include >>
Validate Leave Balance
View contract
Track Attendance

Written Use Case:

Log in Use Case:

Use Case Title: Log in
Primary Actor: Employee, HR Manager
Level: Kite
Stakeholders: Employee, HR Manager
Preconditions: • The user entering to the system.
Minimal Guarantee: Wrong user information's
Succes Guarantee: Login to the account in the system successfully
Trigger: The user access website homepage
Main Success Scenario: 1- fill the user information's (username and password) 2- Send a verification code to the mobile number 3- Login to the System
Extensions: 1.a. The password or username is incorrect 1.a.1. Reset the password 1.a.2. Verify email validity 2.a. The code was not sent to the mobile number 2.a.1. Resend the code 3.a. No internet connection to log into the account 3.a.1. connect to the internet

Submit Leave Request Use Case:

Use Case Title: Submit Leave Request
Primary Actor: Employee
Level: Kite
Stakeholders: Employee
Preconditions: • The employee accesses the submits leave request option in the system.
Minimal Guarantee: leave balance is insufficient
Succes Guarantee: request is submitted to HR
Trigger: The Employee submits a leave request.
Main Success Scenario: 1- Employee selects leave type and dates. 2- System checks available leave balance. 3- If sufficient balance, request is submitted to HR. 4- System sends notification to HR Manager. 5- Employee receives confirmation.
Extensions: 2.a. If leave balance is insufficient 2.a.1. the system denies the request.

Manage Employee Contracts Use Case:

Use Case Title: Manage Employee Contracts
Primary Actor: HR Manager
Level: Kite
Stakeholders: HR Manager
Preconditions: • The HR Manager is logged into the system with the permissions.
Minimal Guarantee: The system doesn't display the employee's contract details.
Succes Guarantee: The contract is successfully created, updated, renewed, or terminated, and the system saves the changes.
Trigger: The HR Manager selects the Manage Contract option in the system.
Main Success Scenario: 1- HR Manager selects an employee profile. 2- The system retrieves the employee's contract details. 3- The HR Manager enters or updates contract details (e.g., start date, end date, salary, role, conditions). 4- The HR Manager submits the changes. 5- The system validates the input and updates the contract record. 6- The system confirms the action and notifies the HR Manager and the employee.
Extensions: 3.a. Employee Has No Existing Contract 3.a.1. The system notifies the HR Manager that no contract is found. 3.a.2. The HR Manager may proceed with creating a new contract. 5.a. System Fails to Update Contract 5.a.1. The system encounters an error while saving the contract changes. 5.a.2. The system notifies the HR Manager and suggests retrying or contacting IT support.

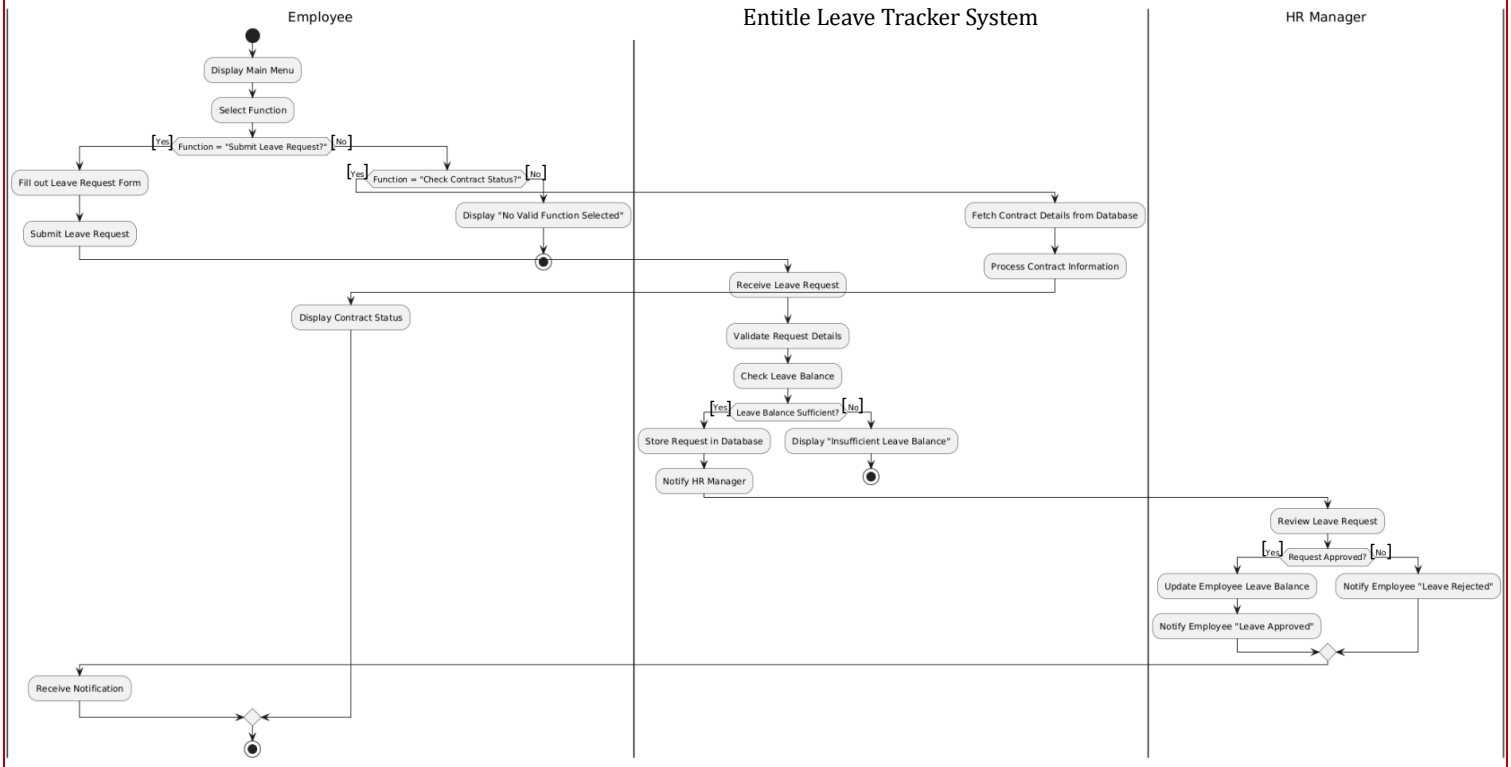
Track Attendance Use Case:

Use Case Title: Track Attendance
Primary Actor: Employee
Level: Kite
Stakeholders: Employee
Preconditions: • Employee enters and logs into the system • System tracks employee attendance data
Minimal Guarantee: System fails to retrieve attendance data
Succes Guarantee: The employee successfully views their attendance details
Trigger: The employee chooses “Track Attendance” option
Main Success Scenario: 1. Employee chooses the Track Attendance section 2. System retrieves the employee’s data 3. System displays the check-in/check-out, working hours, and absence 4. Employee reviews the attendance record
Extensions: 1a. Track attendance section fails to load 1a1. The employee refreshes the page 1a2. The employee reports the issue to HR 2a. Data Retrieval Failure 2a1. The system displays an error message and suggests trying again later. 3a. Attendance Data is Incorrect or Missing 3a1. The employee submits a request to HR to correct the issue. 4a. No Attendance Records Available 4a1. The employee submits a request to HR to correct the issue.

Approve/Reject Leave Use Case:

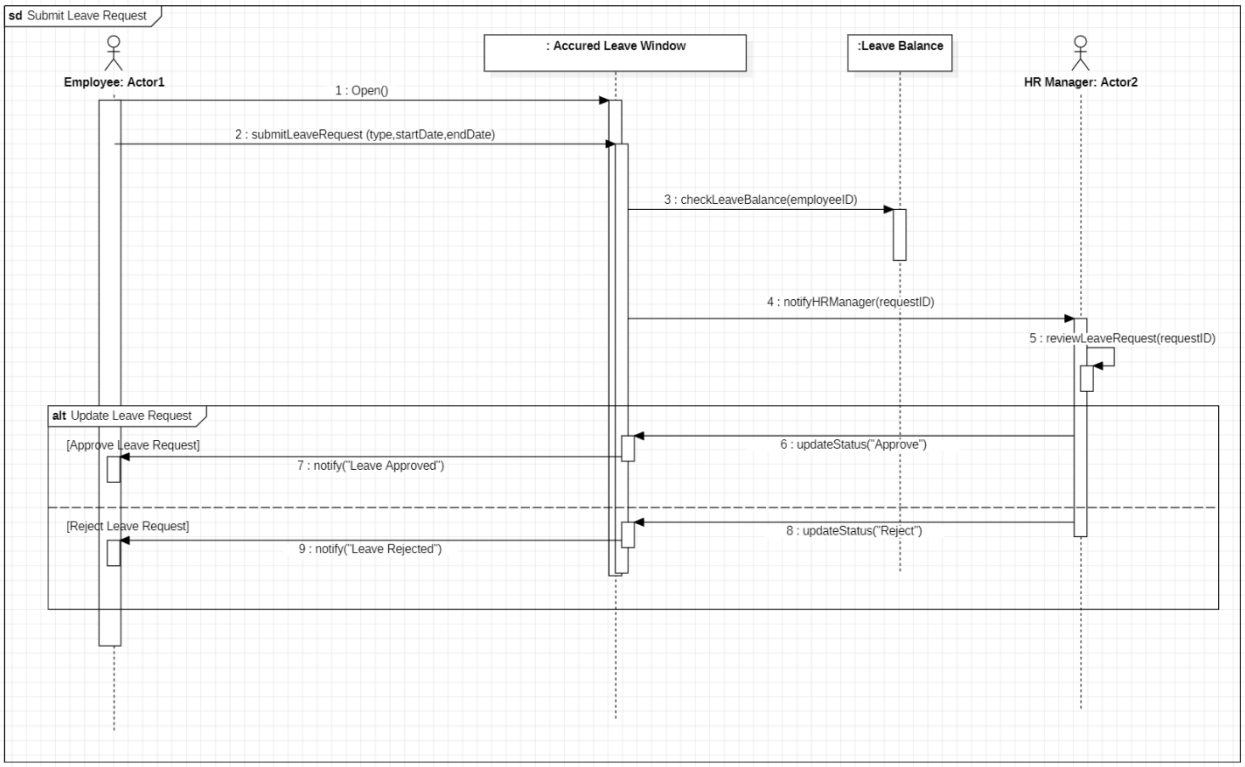
Use Case Title: Approve/Reject Leave
Primary Actor: HR Manager
Level: Kite
Stakeholders: HR Manager
Preconditions: • Employee's leave request is submitted and pending • HR Manager logs in with the authority to approve or reject leave.
Minimal Guarantee: The system can't load pending leave requests
Succes Guarantee: The system updates the pending request status to Approved or Rejected
Trigger: HR Manager chooses the "Leave Requests" section
Main Success Scenario: 1. The system displays a list of pending leave requests of employees 2. The HR Manager selects a request to review its details 3. The HR Manager chooses to Approve or Reject the request 4. The system sends a notification to the employee
Extensions: 1a. Error displaying requests 1a1. The manger refreshes the page or waits 2a. Request information missing 2a1. HR manger contacts the employee for the missing information 3a. Leave status doesn't get updated 3a1. Employees contact the manager for their status 4a. System fails to send the notification 4a1. HR contacts the employee manually 4a2. HR resends the notification

Activity diagram:

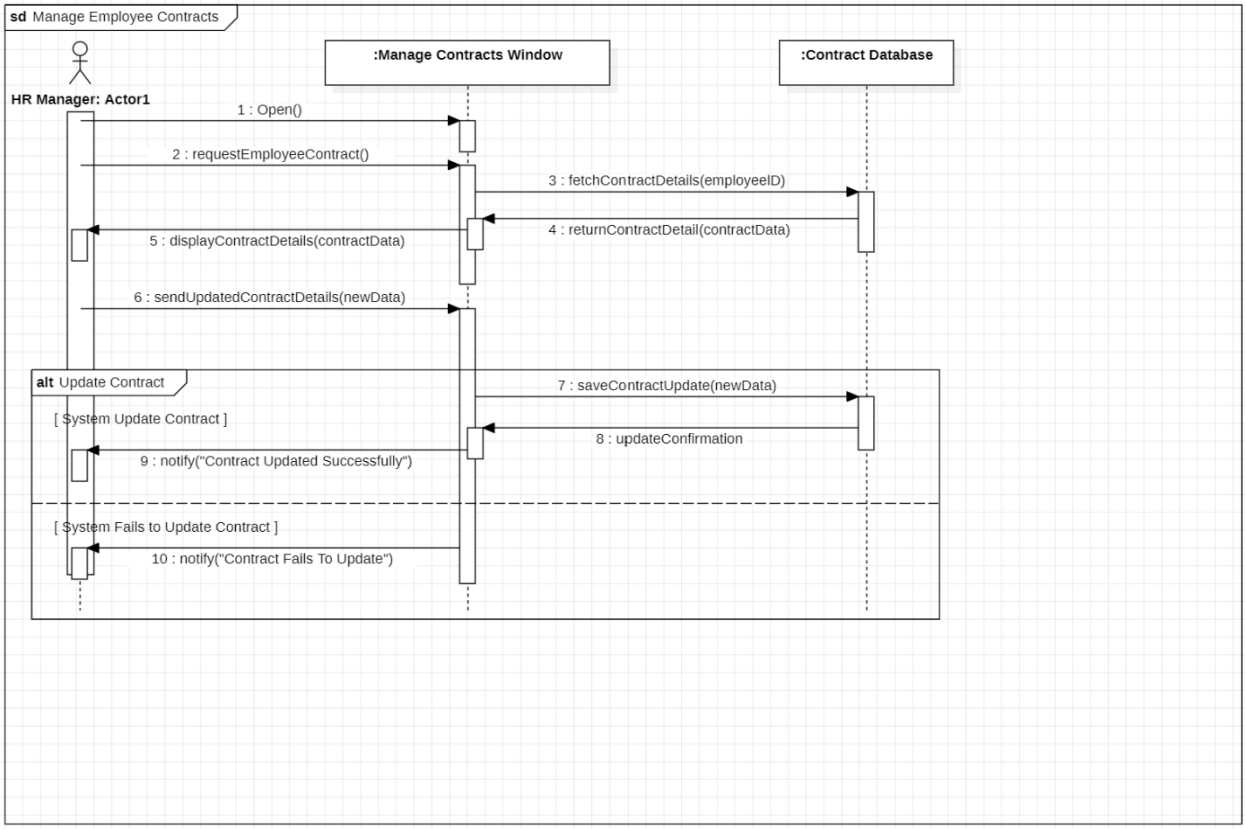


Sequence Diagram:

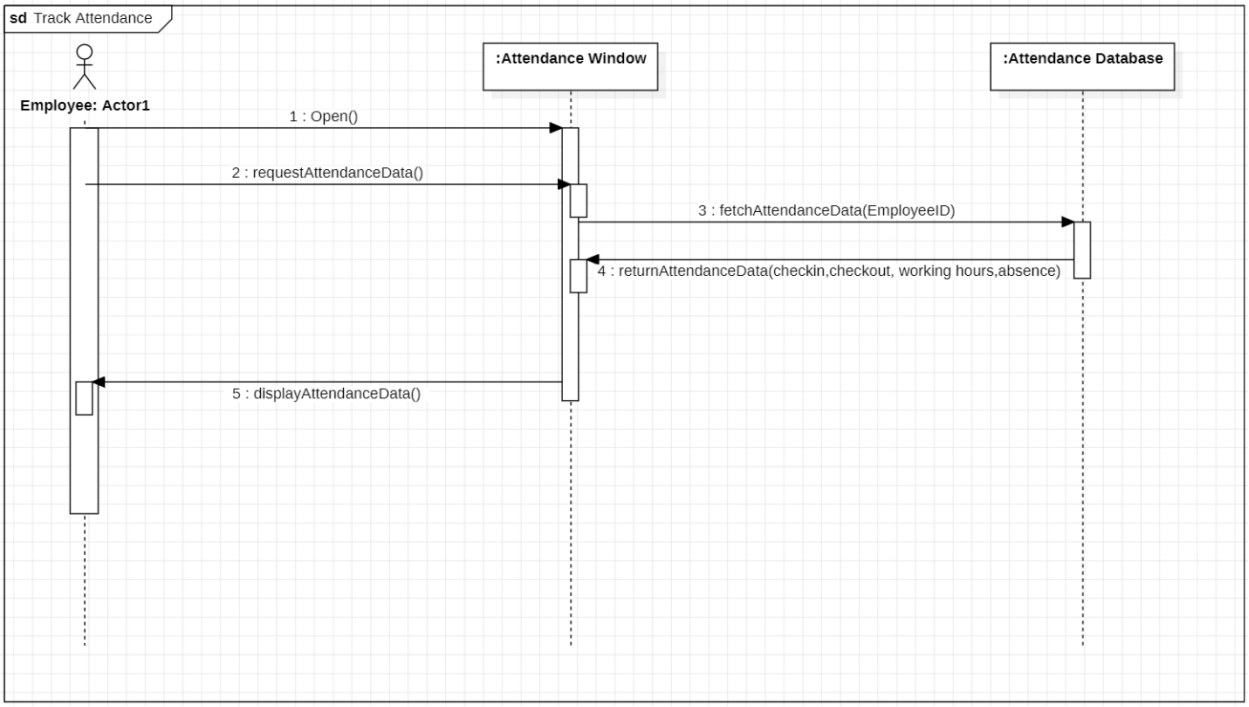
1.Submit Leave Request:



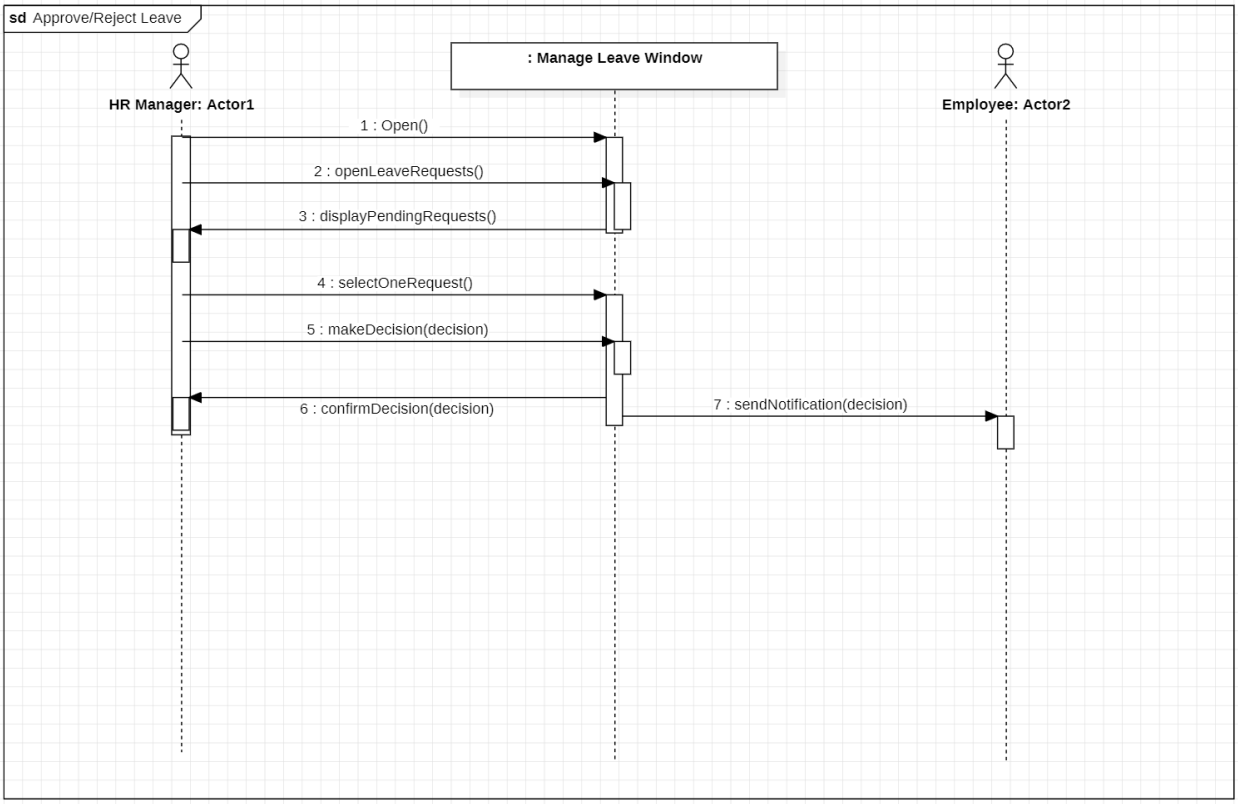
2. Manage Employee Contracts:



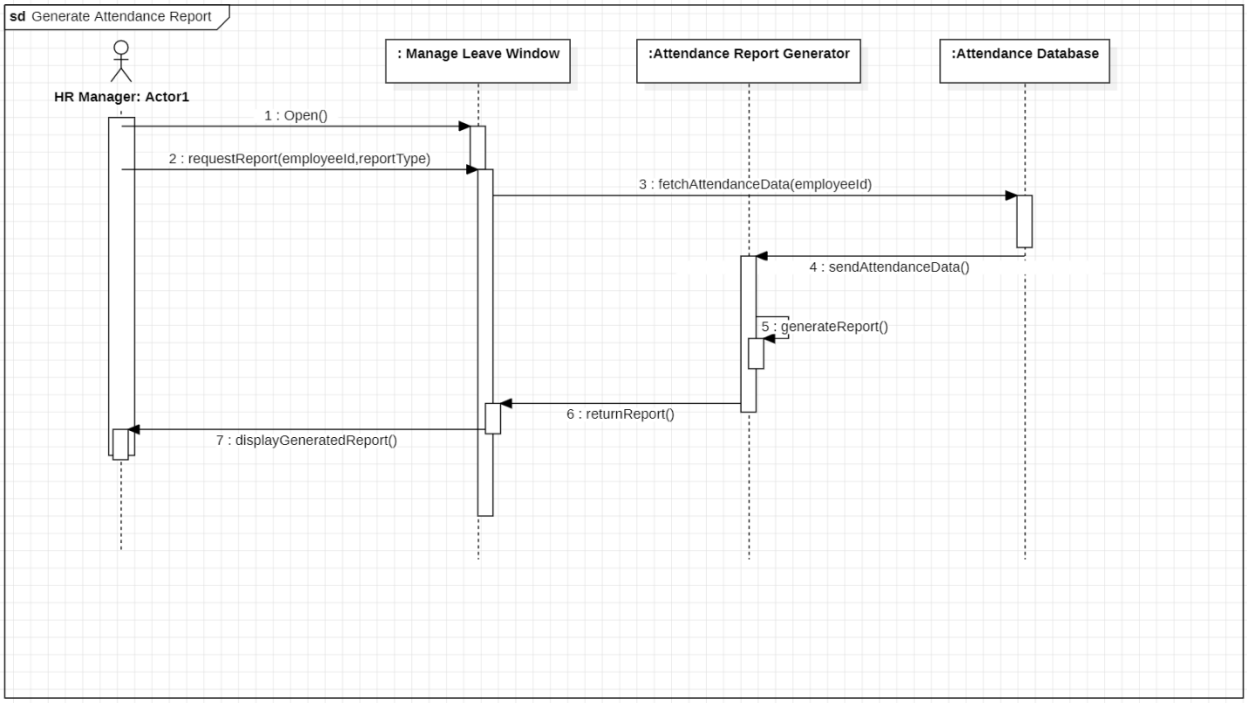
3. Track Attendance:



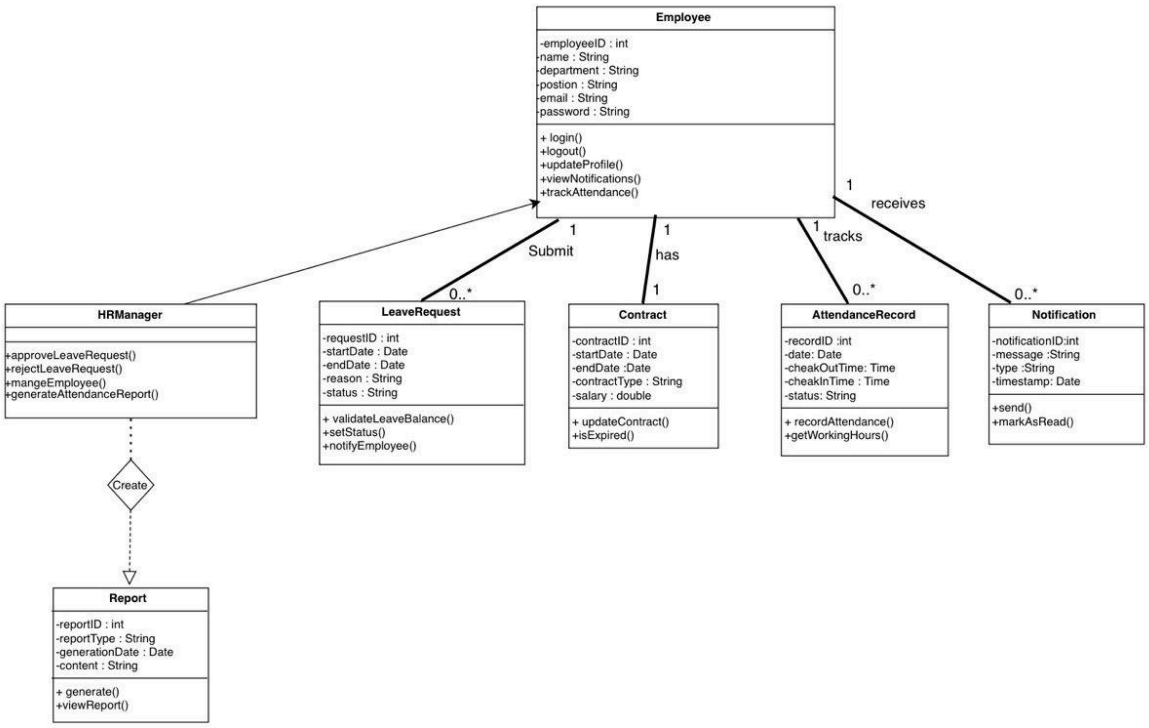
4. Approve/Reject Leave:



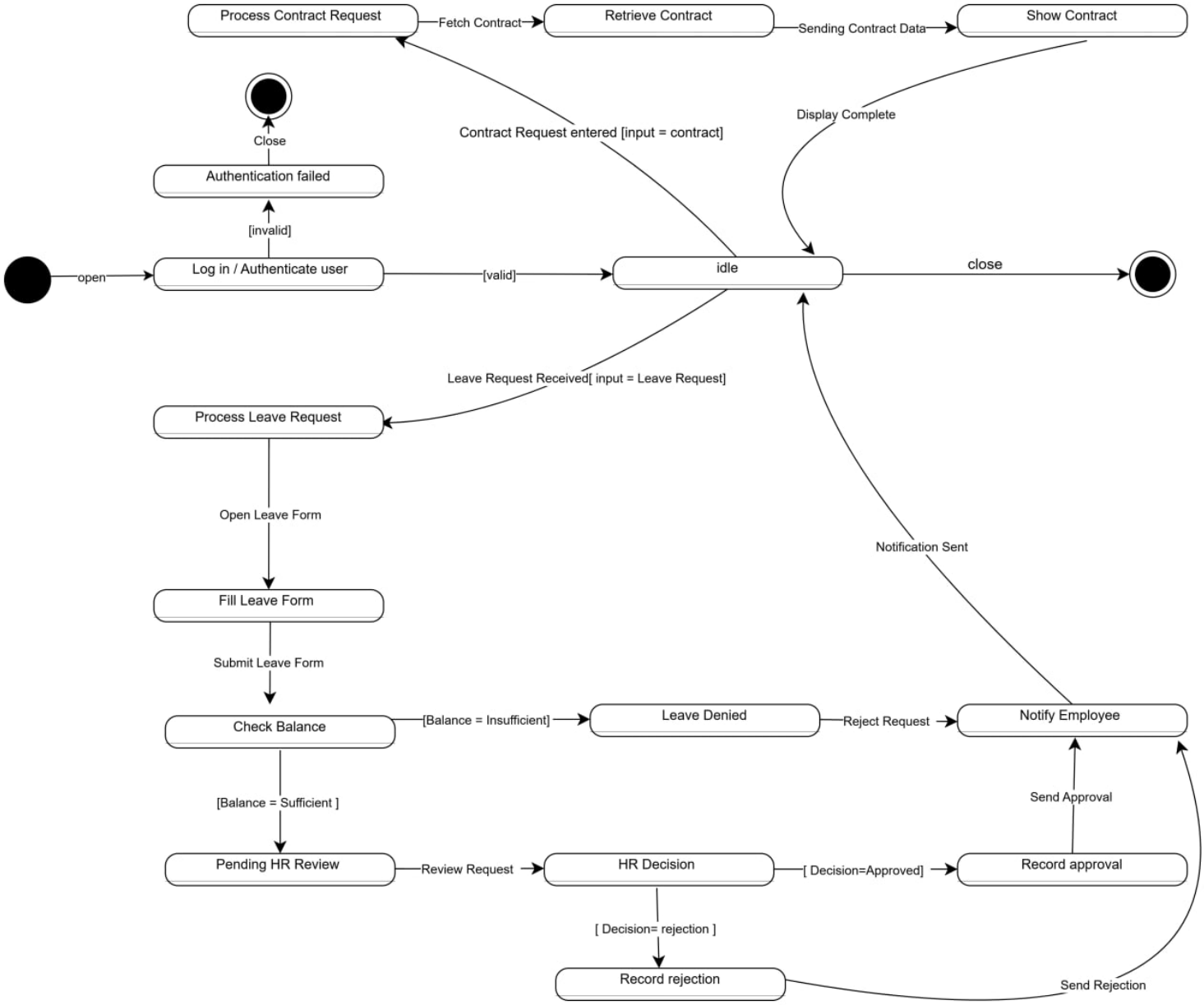
5. Generate Attendance Report:



Class diagram:

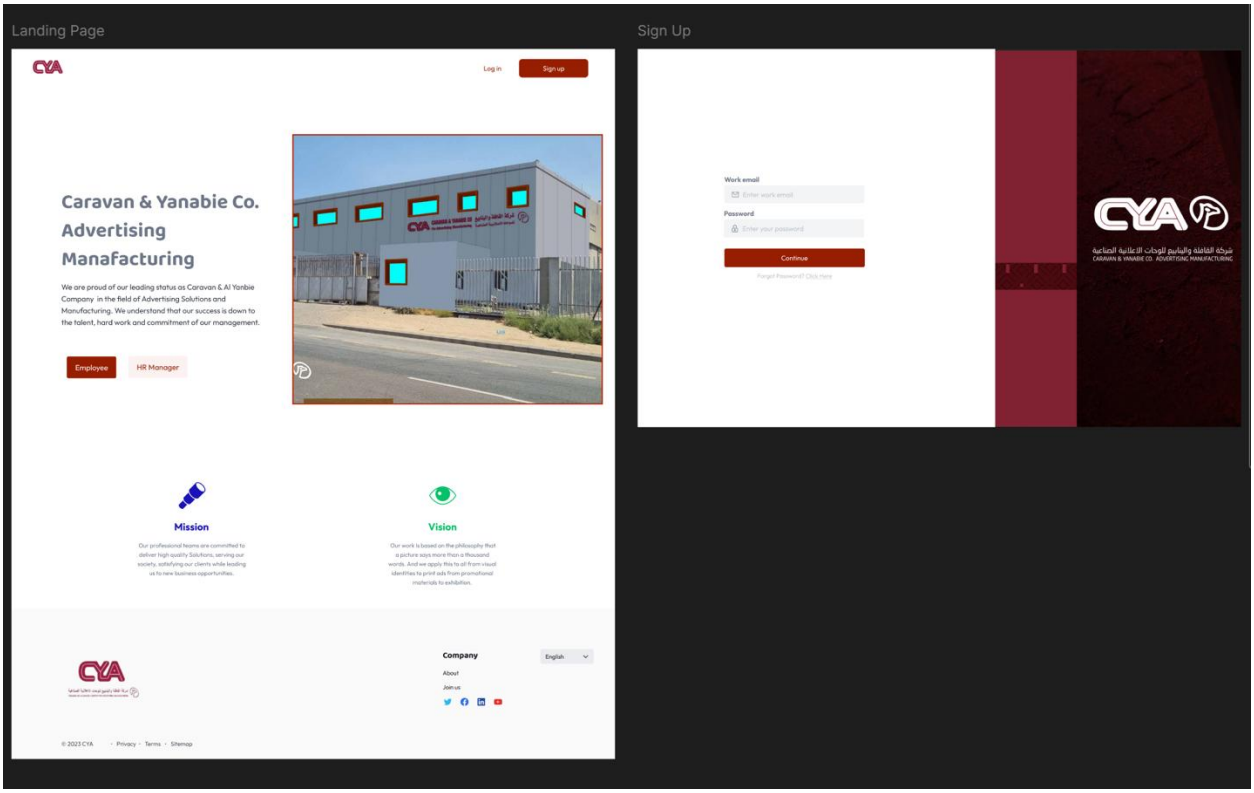


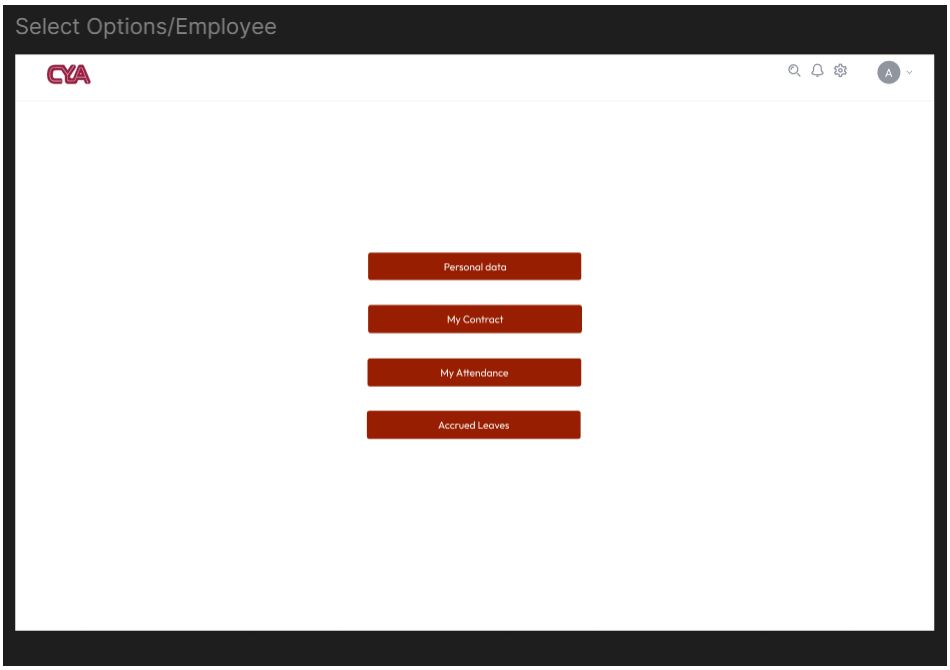
State Diagram:



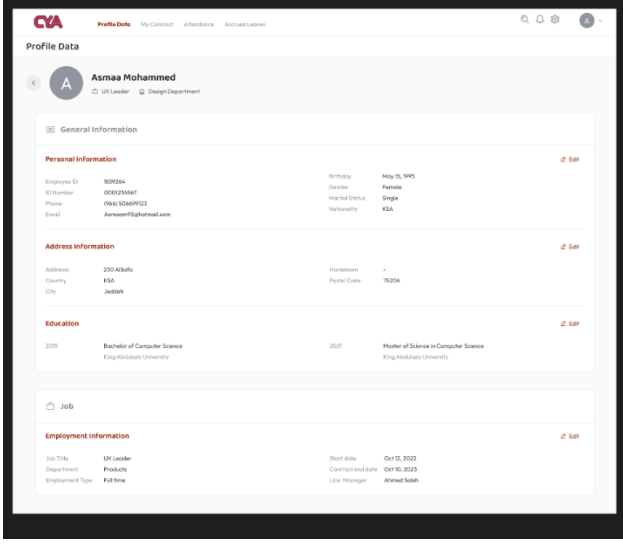
UI Interfaces:

On [Figma]

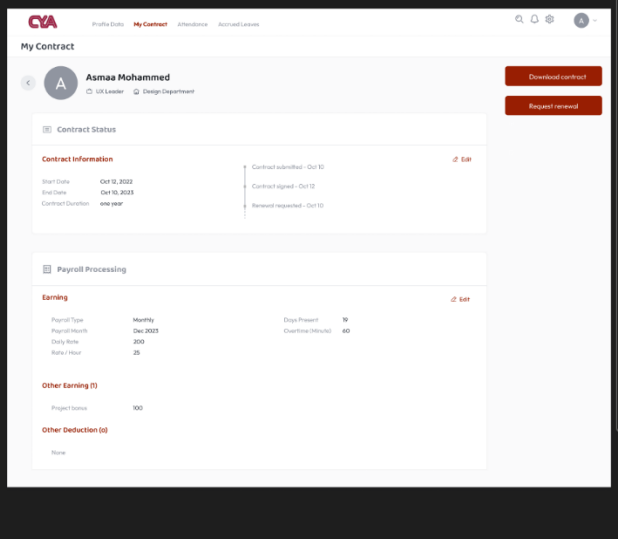




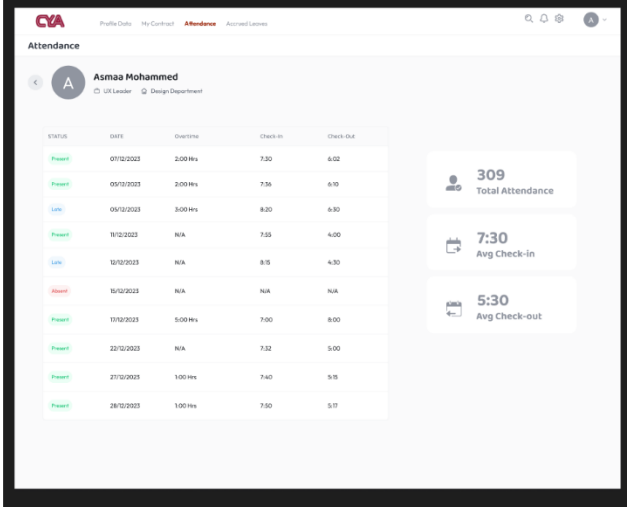
Profile Data/Employee



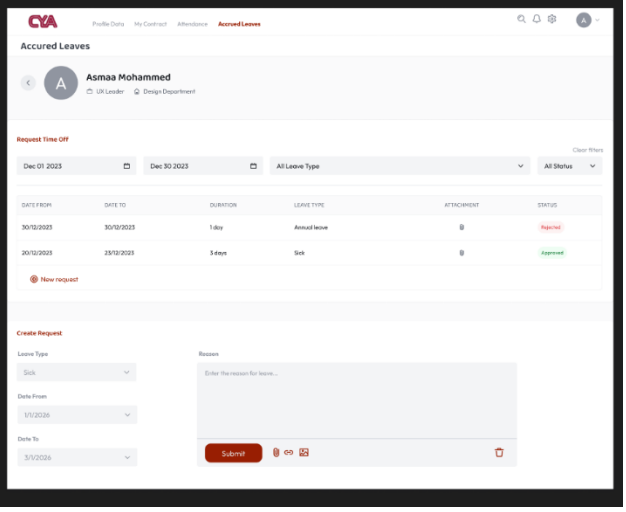
Contract Status/Employee



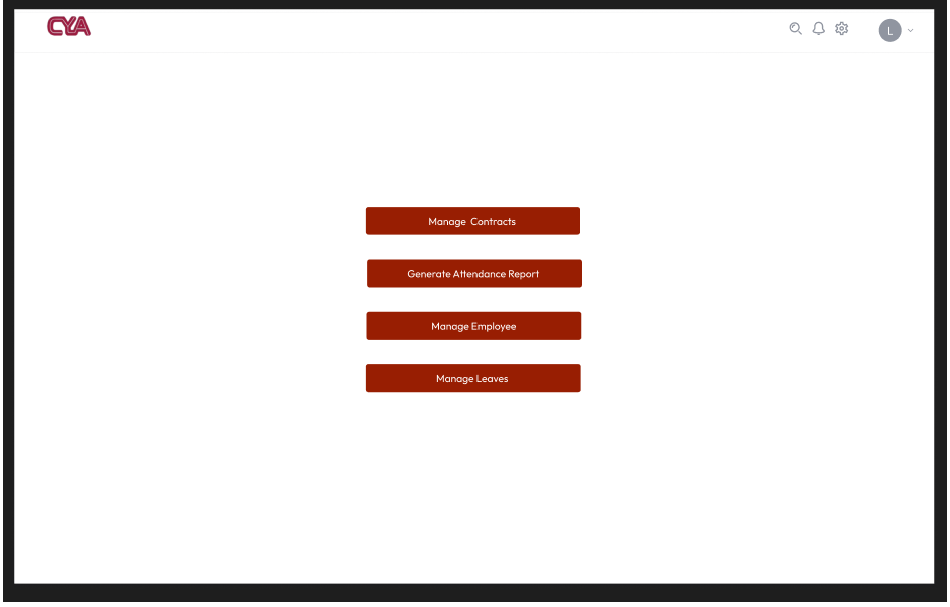
Attendance/Employee



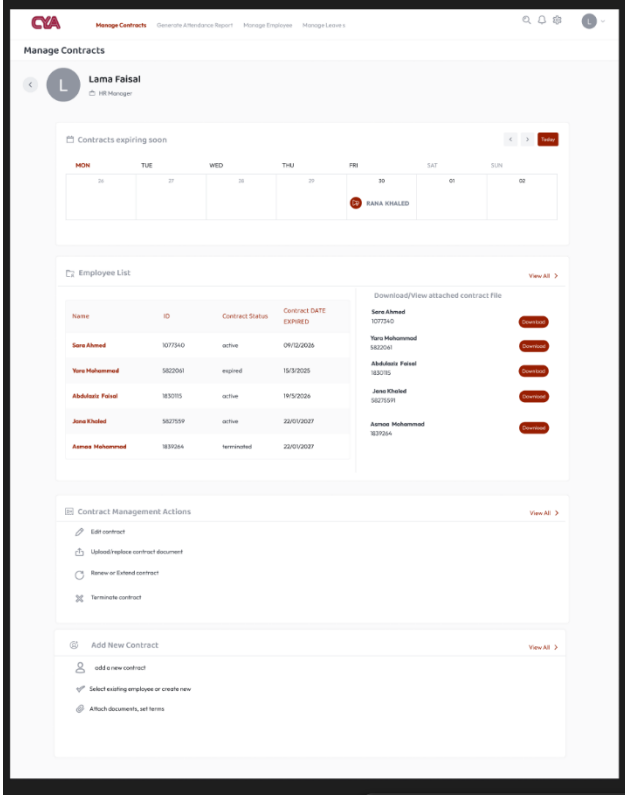
Accured Leaves/Employee



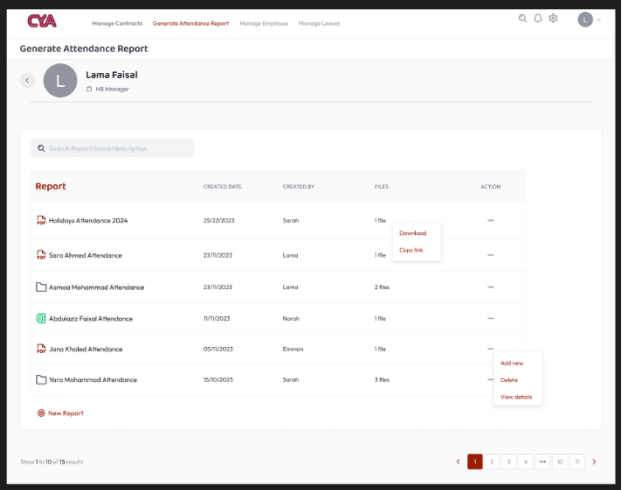
Select Options/Manager



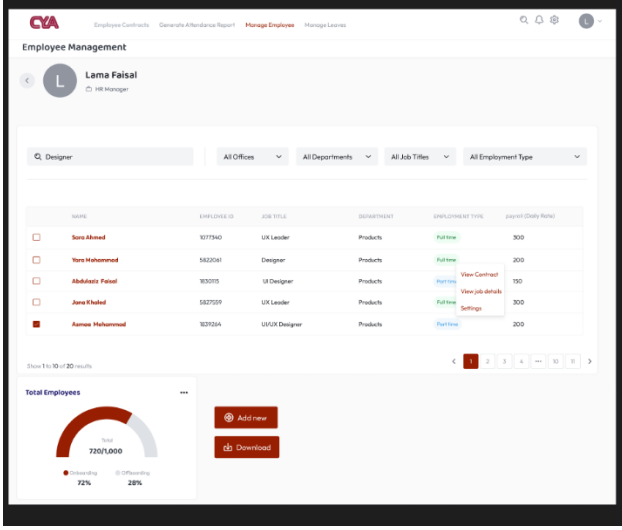
Manage Contracts/Manager



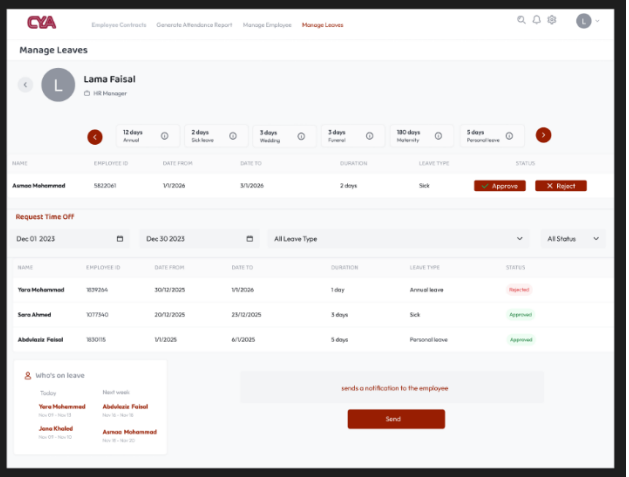
Generate Attendance Report /Manager



Employee Management/Manager



Manage Leave /Manager



Conclusion + Future Direction

In conclusion, the EntitleLeave Tracker System offers real solutions to the everyday problems faced by HR departments. By moving from manual methods to a smart digital platform, the system helps reduce human errors, speeds up processes like leave approval, and ensures better organization of employee records. The system makes it easier for both HR staff and employees to access important information and manage tasks smoothly.

The benefits go beyond just saving time. With better tracking and reporting, HR can now focus on more important strategies like employee development and performance improvement. The system also builds stronger communication by sending automatic notifications for contracts, leave, and attendance issues.

Looking to the future, we plan to expand the system by creating a mobile app version so users can access it anytime, from anywhere. We also want to integrate it with official government platforms for tasks like employee verification and residency renewal. Additionally, we hope to include smart data analytics to help managers understand employee trends and make better decisions.

Our goal is to continue improving this system so it becomes a full HR solution that supports every part of the employee journey in a modern and digital way.

References

- Interview with HR: Haneen Almansouri
- Figma Interface Design
- CPIT250 Course Material
- System Design Principles – IBM Docs

Group Work Table

TARAFH ALAKOR	1- Interview Outline 2- Questions 3- Data Flow Diagram (DFD) (Context, Level 0) 4- Written Use Case (LogIn) 5- Sequence Diagram (Approve/Reject Leave) 6- Activity Diagram 7- State Diagram
DANYAH BARASHEED	1- Problem Description 2- Solution Description 3- Data Flow Diagram (DFD) (Level 0) 4- Written Use Case (Approve/Reject Leave) 5- Sequence Diagram (Generate Attendance Report) 6- UI Interfaces
GHAIDA HADI	1- Architecture diagram 2- Use Case 3- Written Use Case (Submit Leave Request) 4- Sequence Diagram (Submit Leave Request) 5- Class diagram 6- Conclusion + Future Direction
MARYHAN ABUALELA	1- Introduction 2- Interview main result 3- Written Use Case (Manage Employee Contracts) 4- Sequence Diagram (Manage Employee Contracts) 5- Class diagram 6- UI Interfaces
AMAL ALJOHANI	1- List of requirements 2- Data Flow Diagram (DFD) (Level 1, Level 2) 3- Written Use Case (Track Attendance) 4- Sequence Diagram (Track Attendance) 5- Activity Diagram 6- State Diagram